

Estimating the Causal Effects of a Growth Mindset Intervention on Academic Performance

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Introduction / Motivation

- Transitioning to high school is particularly difficult for lower-performing students.
- The National Study of Learning Mindsets (Yeager et al., 2019) tested a brief online intervention promoting a growth mindset—the belief that intelligence can grow with effort.
- The intervention led to modest but meaningful GPA gains and increased enrollment in advanced math among lower-achieving students.
- This project uses a synthetic dataset modeled on that study to estimate the average treatment effect (ATE) and assess how the intervention's impact may vary across students and school contexts.

Data Exploration

- Dataset includes 10,391 students from 76 schools.
- Each record contains:
 - Standardized outcome ('y') and treatment indicator ('z')
 - Demographics: gender, race, first-generation status
 - Academic variables: test scores and mindset
 - School-level context: poverty rate, racial makeup, size
- About one-third of students received the intervention ('z = 1').
- Outcome 'y' is centered near zero with a slight positive skew—most students scored around average, with some high outliers.
- Gender distribution is nearly even; 63% are first-generation students; race spans 15 categories.
- Mindset and test scores show high variability, indicating diverse academic backgrounds.

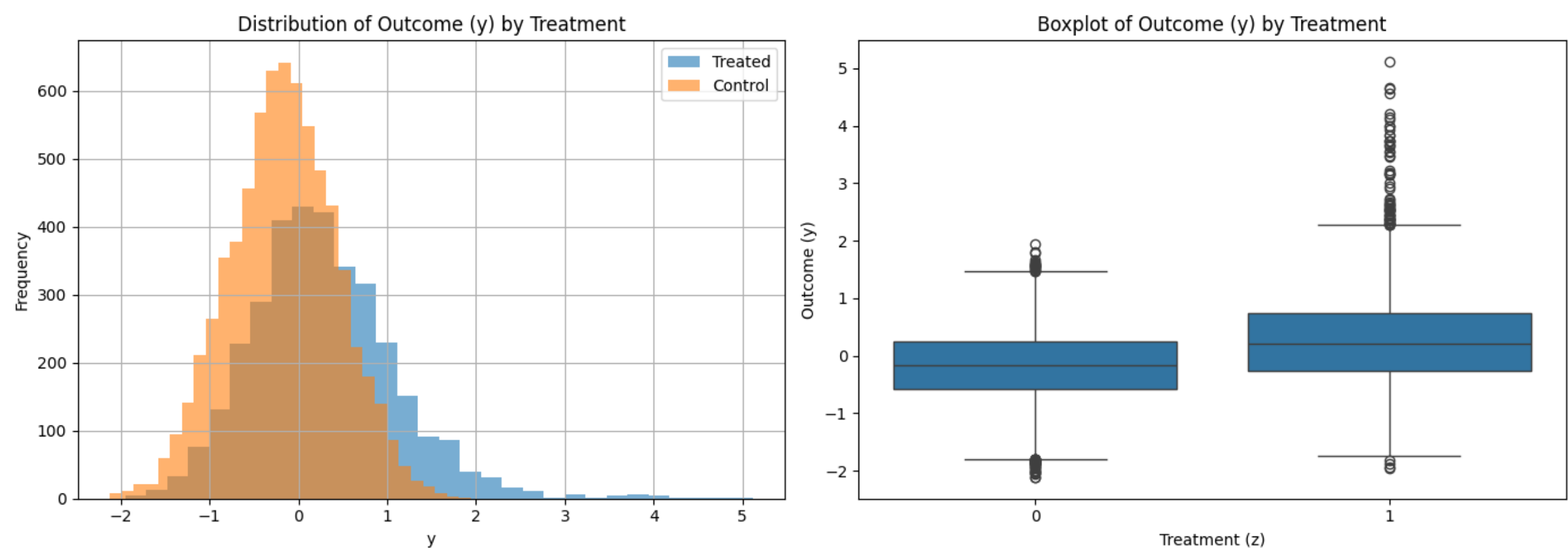


Figure 1. Distribution and boxplot of outcome variable y by treatment group.

Substantial variation across schools (in size, poverty, and racial makeup) highlights the importance of accounting for context in analysis. Initial visualizations show that treated students tend to score slightly higher than controls, with more positive outcomes and more high-end outliers. This suggests possible heterogeneity in the treatment effect.

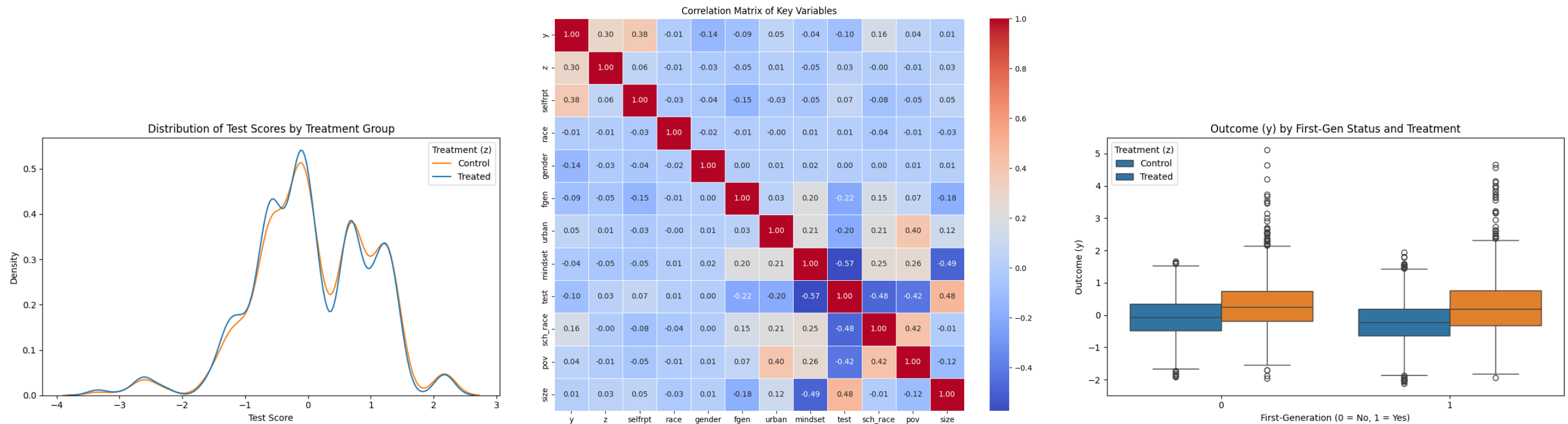


Figure A. Test score distributions by treatment group show strong overlap, indicating good baseline comparability.

Figure B. Correlation matrix reveals modest associations among covariates, with test and mindset moderately correlated.

Figure C. Boxplot of outcomes shows a treatment effect across both first-gen and non-first-gen students.

Data Analysis: Methods and Assumptions

I use two main methods in my project:

1. Regression Adjustment

I fit a model that predicts a student's outcome based on whether they received the intervention and on background characteristics like test scores, mindset, gender, and race. This helps me adjust for existing differences between treated and control students when estimating the treatment effect.

2. Stratification

I partition students into subgroups according to a key variable, such as their test scores, and compare treated and control students within each subgroup. I then compute the mean of these differences to get an overall estimate. This approach does not involve a particular model and enables an analysis of how the effect can differ across various categories of students.

The selection of these covariates was based on their strong theoretical and empirical connections with academic attainment. Additional variables, including school-level variables, were excluded from the primary model in the interests of parsimony; however, they were subsequently examined employing fixed effects and stratified analyses.

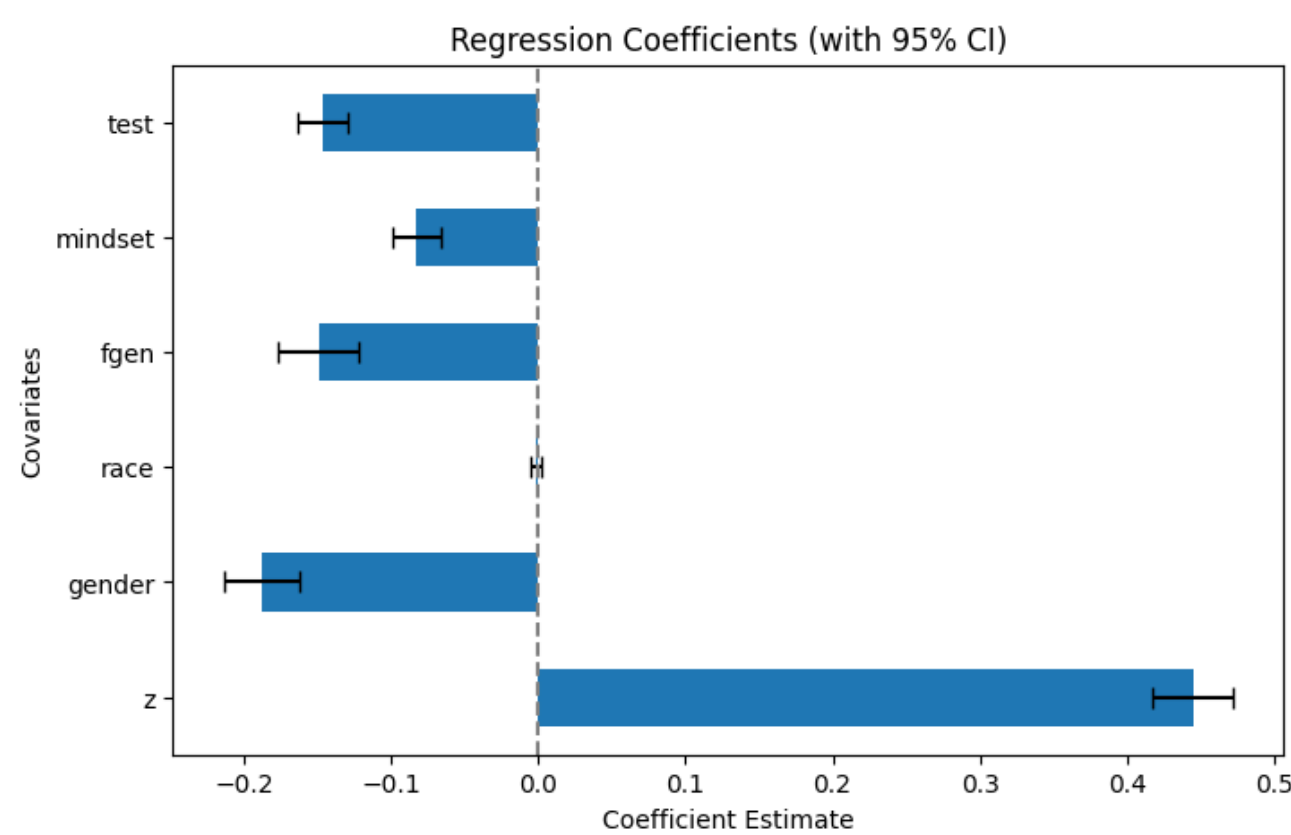


Figure 2. Regression coefficients with 95% CI. Treatment is significant.

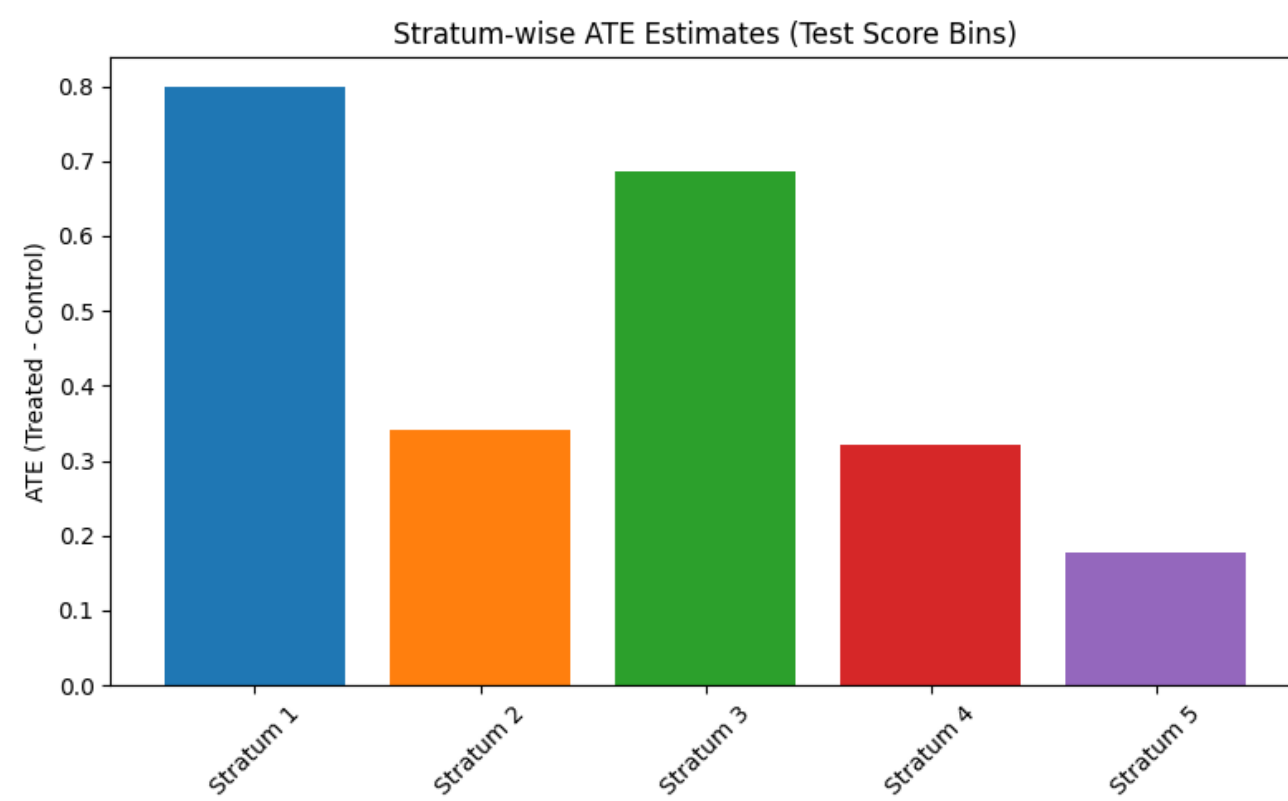


Figure 3. Stratified ATEs. Stronger effects in lower performers.

Additional Results:

- Figure 4:** Regression coefficients from a model with interaction terms. The treatment effect remains positive, with its strength varying by prior test score and first-generation status.
- Figure 5:** Standardized mean differences (SMDs) between treated and control groups. All covariates fall within ± 0.1 , showing good pre-treatment balance.
- Figure 6:** Estimated treatment effects from separate regressions by test score stratum. The intervention appears most beneficial for lower-performing students.

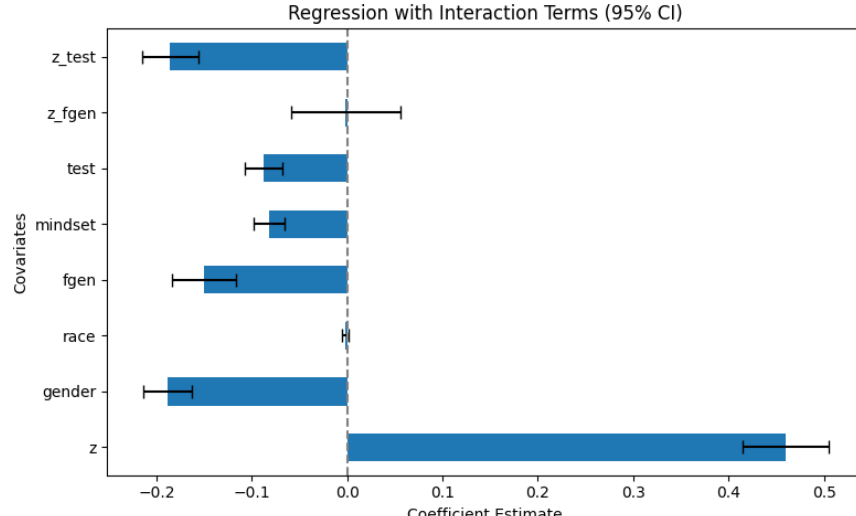


Figure 4. Regression with interaction terms.

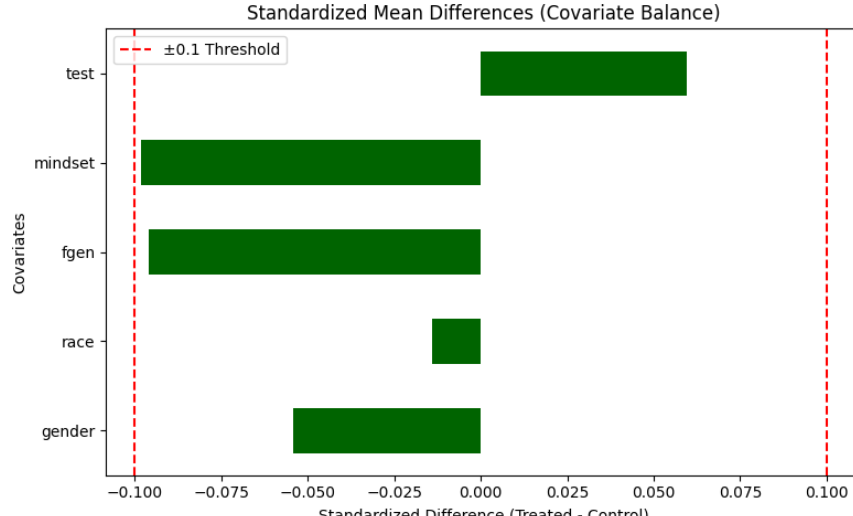


Figure 5. Covariate balance.

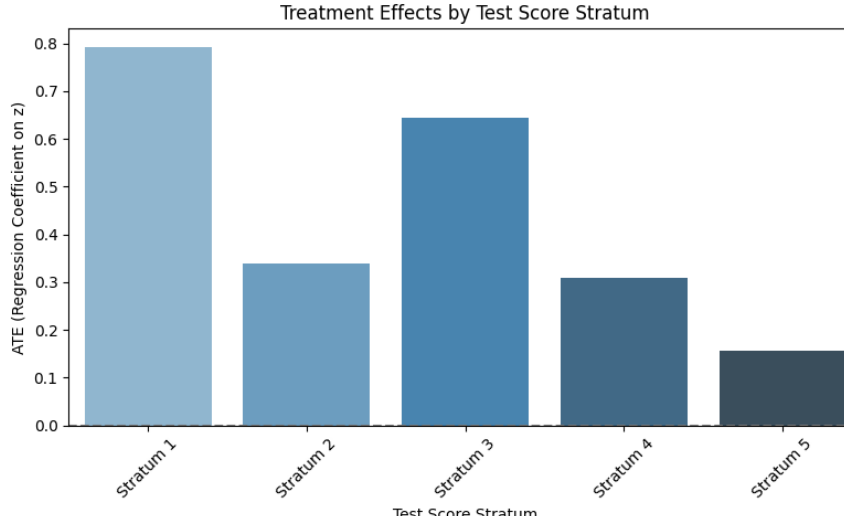


Figure 6. Treatment effect by test score stratum.

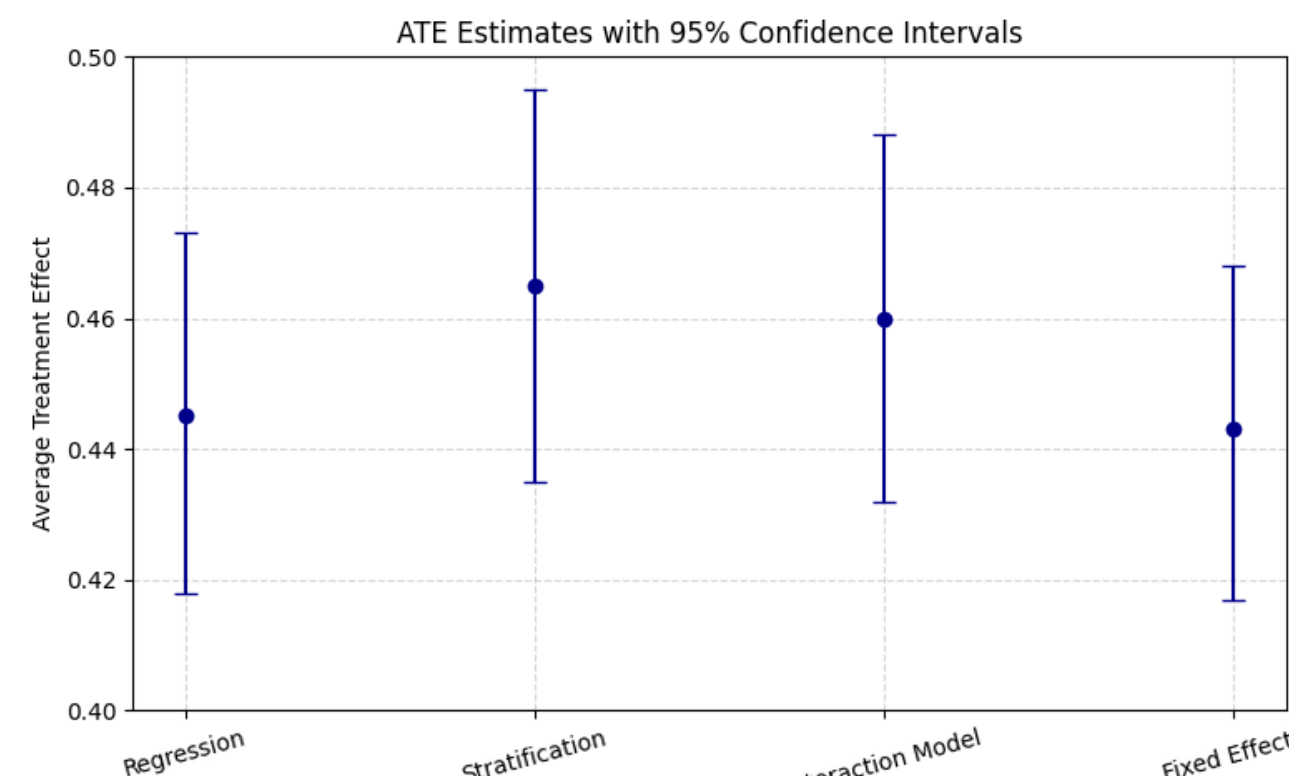
Assumptions

Both methods rely on a few core assumptions: (1) *Conditional ignorability*—after adjusting for observed variables, treatment is assumed to be as good as randomly assigned; (2) *Positivity*—all students have a non-zero chance of receiving treatment or control; (3) *No unmeasured confounding*—all relevant covariates have been included; and (4) *SUTVA*—the treatment effect for one student does not depend on others.

Results

The regression model estimates an ATE of 0.445 with a 95% CI of (0.418, 0.473). Interaction terms show stronger effects for first-gen and lower-scoring students, confirming treatment heterogeneity.

Covariate balance is supported by all standardized mean differences being below ± 0.1 :



Covariate	SMD (Treated - Control)
Gender	-0.054
Race	-0.014
First-Gen	-0.096
Mindset	-0.098
Test Score	0.060

Table 1. Standardized mean differences for key covariates. All values < 0.1 .

Figure 7. ATE estimates with 95% confidence intervals across methods.

Stratified analysis shows the largest effect in the lowest-scoring group (0.79), declining across higher test score groups. This consistency across methods supports a robust, positive effect, particularly for students most at risk.

In the interaction model (Fig. 4), the coefficient for **z_test** was -0.20 (95% CI: -0.24, -0.16), and for **z_fgen** it was -0.08 (95% CI: -0.14, -0.02), indicating a stronger treatment effect for lower-scoring and first-generation students. The main treatment effect ('z') remained positive and significant.

Discussion

While the analysis controls for key observed variables, the synthetic nature of the dataset means it may not capture real-world sources of unmeasured confounding. Variables such as student motivation, classroom environment, or teacher effects could influence outcomes, but are not represented here. This limits the generalizability of causal conclusions to real-world contexts.

Finally, if peer interactions affect results, the SUTVA might not hold. For instance, if students gain indirect advantages from their peers who were given the intervention. Despite these drawbacks, there is some comfort in knowing that the estimated treatment effect is fairly robust due to the consistency of results across methods.

Incorporating Peer Feedback

Several improvements in this final version were informed by insightful peer reviews. In response to feedback about model simplification, I clarified my covariate selection and acknowledged the value of including additional variables in future extensions. To address concerns about text density, I revised and condensed several sections for clarity and visual balance. I also expanded the discussion of limitations to explicitly address unobserved confounding and the synthetic nature of the dataset. Finally, based on suggestions to better quantify heterogeneity, I added numeric summaries of interaction effects and subgroup ATEs to improve interpretability.

References

- Yeager, D. S., Hanselman, P., Walton, G. M., et al. (2019). A national experiment reveals where a growth mindset improves achievement. *Nature*, 573(7774), 364–369.